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Linux Containers and Kernel Sharing

In simple terms, Linux containers are a lightweight alternative to virtualization.

A virtual machine contains and runs the entire guest operating system in a virtualized environment.

The virtual machine, in turn, runs on top of an environment such as a hypervisor that manages access to the physical resources of the host system.

Container Uses and Advantages

The main advantage of containers is that they require considerably less resource overhead than virtualization allowing many container instances to be run simultaneously on a single server.

They can be started and stopped rapidly and efficiently in response to demand levels.

In addition, containers run natively on the host system providing a level of performance that a virtual machine cannot match.

Containers are also highly portable and can be easily migrated between systems. Combined with a container management system such as Docker, OpenShift, and Kubernetes,

it is possible to deploy and manage containers on a vast scale spanning multiple servers and cloud platforms, potentially running thousands of containers.

Containers are frequently used to create lightweight execution environments for applications.

In this scenario, each container provides an isolated environment containing the application together with all of the runtime and supporting files required by that application to run.

The container can then be deployed to any other compatible host system that supports container execution and runs without any concerns that the target system may not have the necessary runtime configuration for the application - all of the application's dependencies are already in the container.

Working with Containers on CentOS Stream 9

Before starting with containers, the first step is to install all of the container tools using the following command:

dnf install container-tools

 # podman pull quay.io/centos/centos:stream9-minimal

 # podman images

podman inspect quay.io/centos/centos:stream9-minimal

root@centos9 ~]# # podman images

```
[root@centos9 ~]# podman run --rm quay.io/centos/centos:stream9-minimal cat
/etc/passwd
root:x:0:0:root:/root:/bin/bash
bin:x:1:1:bin:/bin:/sbin/nologin
daemon:x:2:2:daemon:/sbin:/sbin/nologin
adm:x:3:4:adm:/var/adm:/sbin/nologin
lp:x:4:7:lp:/var/spool/lpd:/sbin/nologin
sync:x:5:0:sync:/sbin:/bin/sync
shutdown:x:6:0:shutdown:/sbin:/sbin/shutdown
halt:x:7:0:halt:/sbin:/sbin/halt
mail:x:8:12:mail:/var/spool/mail:/sbin/nologin
operator:x:11:0:operator:/root:/sbin/nologin
games:x:12:100:games:/usr/games:/sbin/nologin
ftp:x:14:50:FTP User:/var/ftp:/sbin/nologin
nobody:x:65534:65534:Kernel Overflow User:/:/sbin/nologin
[root@centos9 ~]#
```

 # podman run --name=mycontainer -it quay.io/centos/centos:stream9-minimal /bin/bash

```
[root@centos9 ~]# podman ps -a
CONTAINER ID  IMAGE                                  COMMAND     CREATED
STATUS                  PORTS                       NAMES
7fb97fe842ed  docker.io/ollama/ollama:latest          serve       29 hours ago
Exited (0) 29 hours ago   0.0.0.0:11434->11434/tcp  ollama
a4a7769f1ce2  quay.io/centos/centos:stream9-minimal  /bin/       5 hours ago
Created                                             mycontainer
52d8661ab3b0  quay.io/centos/centos:stream9-minimal  /bin/bash   5 hours ago   Up 5
hours                                          mycontainer2
cf6fdbe7b472  localhost/my-coreutils-image:latest    bash        4 hours ago
Exited (127) 4 hours ago                            laughing_chatelet
d4abe80a2e1a  localhost/my-coreutils-image:latest    bash        4 hours ago
Exited (0) 5 minutes ago                            epic_jang
```

podman run --name=mycontainer4 -it quay.io/centos/centos:stream9-minimal /bin/bash
bash-5.1#

TTHE PREVIOUS PROMPT IS FROM THE NEW CONTAINER

IN ANOTHER TERMINAL SESSION

[bobb@centos9 ~]$ podman ps -a

```
[bobb@centos9 ~]$ podman ps -a
CONTAINER ID  IMAGE                                  COMMAND     CREATED
STATUS                   PORTS                       NAMES
7fb97fe842ed  docker.io/ollama/ollama:latest          serve       30 hours ago
Exited (0) 29 hours ago    0.0.0.0:11434->11434/tcp  ollama
```

```
a4a7769f1ce2  quay.io/centos/centos:stream9-minimal  /bin/       5 hours ago
Created                                              mycontainer
52d8661ab3b0  quay.io/centos/centos:stream9-minimal  /bin/bash   5 hours ago    Up
5 hours                                           mycontainer2
cf6fdbe7b472  localhost/my-coreutils-image:latest    bash        5 hours ago
Exited (127) 5 hours ago                             laughing_chatelet
d4abe80a2e1a  localhost/my-coreutils-image:latest    bash        4 hours ago
Exited (0) 23 minutes ago                            epic_jang
cd257bdb2571  quay.io/centos/centos:stream9-minimal  /bin/bash   5 minutes ago
Exited (0) 4 minutes ago                             mycontainer4
b5a62da4ecff  quay.io/centos/centos:stream9-minimal  /bin/bash   4 minutes ago  Up
4 minutes                                         mycontainer5
[bobb@centos9 ~]$

[bobb@centos9 ~]$ podman exec -it mycontainer5 /bin/bash
bash-5.1#

bash-5.1# pwd
/
bash-5.1# exit
exit
[bobb@centos9 ~]$ podman attach mycontainer5

bash-5.1#
```

podman start mycontainer5

```
[bobb@centos9 ~]$ podman ps -a
CONTAINER ID  IMAGE                                  COMMAND     CREATED
STATUS                     PORTS                     NAMES
...
b5a62da4ecff  quay.io/centos/centos:stream9-minimal  /bin/bash   14 minutes ago  Up
3 seconds                                         mycontainer5
[bobb@centos9 ~]$

[bobb@centos9 ~]$ podman exec -it mycontainer5 /bin/bash
bash-5.1#
```

IN ANOTHER TERMINAL SESSION

 # podman pause mycontainer5

podman ps -a

```
CONTAINER ID  IMAGE                                  COMMAND     CREATED
STATUS                     PORTS                     NAMES
…
b5a62da4ecff  quay.io/centos/centos:stream9-minimal  /bin/bash   24 minutes ago
Paused                                               mycontainer5
```

IN THE CONTAINER IN THE OTHER TERMINL SESSION YOU CAN'T TYPE A COMMAND

podman unpause mycontainer5

YOU CAN NOW TYPE IN COMMANDS IN THE OTHER TERMINAL SESSION

Running Ollama inside a Podman container is not just a “nice-to-have”—it solves several real engineering problems that show up in multi-device GPU setups like yours (Jetson Orin ↔ Ubuntu ↔ Mac Mini). The short answer:

You containerize Ollama to gain isolation, reproducibility, portability, and clean GPU/runtime control.

```
[bobb@centos9 ~]$ sudo dnf install podman -y
[sudo] password for bobb:
Last metadata expiration check: 0:09:05 ago on Wed 05 Aug 2026 03:42:00 PM EDT.
Package podman-6:5.8.5-1.el9.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!

[bobb@centos9 ~]$ podman --version

podman version 5.8.5

[bobb@centos9 ~]$ podman volume create ollama_storage
ollama_storage
```

```
[bobb@centos9 ~]$ podman pull docker://docker.io/ollama/ollama:latest
Trying to pull docker.io/ollama/ollama:latest...
Getting image source signatures
Copying blob 1d42b9ea7560 done   |
Copying blob 733602a186b4 done   |
Copying blob 966c395d29cb done   |
Copying blob 7d1c8af65e66 done   |
Copying config cdcad7aac0 done   |
Writing manifest to image destination
cdcad7aac04c5201de7de9e6962cba0f2c72d51a0971a48127cf3c401e30157c

[bobb@centos9 ~]$ podman run -d \
-v ollama_storage:/root/.ollama \
-p 11434:11434 \
--name ollama \
ollama/ollama
90f10794a50d4c5da481964b660f14dfd6c9e27302130faaedc62124c9c92fd8

[bobb@centos9 ~]$ podman exec -it ollama ollama run llama3
pulling manifest
pulling 6a0746a1ec1a: 100%
 ▕████████████████████████████████████████████
████████████████████████████████▏ 4.7 GB/4.7 GB  108
MB/s      0s
verifying sha256 digest
writing manifest
success
>>> tell me a story about a robot that learns to paint
What an intriguing request! Here's a tale about a robot named Robby, who discovers
the joy of painting:

In the year 2050, a team of brilliant engineers at TechCorp created Robby, a
```

```
cutting-edge humanoid robot designed for assembly-line work. Robby's creators
believed that by giving him advanced
sensors and processing power, he could learn to perform complex tasks with
precision and speed...
...


>>> /bye

[bobb@centos9 ~]$ podman ps
CONTAINER ID  IMAGE                            COMMAND     CREATED         STATUS
PORTS                     NAMES
90f10794a50d  docker.io/ollama/ollama:latest  serve       13 minutes ago  Up 13
minutes  0.0.0.0:11434->11434/tcp  ollama

[bobb@centos9 ~]$ curl http://localhost:11434/api/generate -d '{
"model": "llama3",
"prompt": "Why use containers?",
"stream": false
}'
```

{"model":"llama3","created_at":"2026-08-
05T20:37:57.949319319Z","response":"Containers, also known as containerization,
have become increasingly popular in recent years, and for good reason...
...

```
[bobb@centos9 ~]$
```

```
podman exec -it ollama ollama run llama3
```

It executes a command inside a running Podman container named ollama, opens an interactive shell, and then runs the Llama-3 model through Ollama inside that container.

🔍 Breakdown of each part

1. podman exec

Runs a command inside an already-running container.

2. -it

- -i → interactive (keeps STDIN open)
- -t → allocate a pseudo-TTY

This lets you talk to the model interactively.

3. Container name: ollama

This tells Podman which container to enter.
4. Command inside the container: ollama run llama3

Runs the Llama-3 model using the Ollama runtime installed inside the container.

If the model isn't already pulled, Ollama will download it first.
🧠 What actually happens on your system

Podman attaches your terminal to the container.

Inside the container, it launches the Ollama CLI.

Ollama loads the llama3 model.

You get an interactive prompt where you can type messages to the model.

Creating a container from scratch

[bobb@centos9 ~]$ **buildah from ubuntu:22.04**

ubuntu-working-container-3

[bobb@centos9 ~]$ **buildah mount ubuntu-working-container**

Error: cannot mount using driver overlay in rootless mode. You need to run it in a `buildah unshare` session

[bobb@centos9 ~]$ **buildah unshare**

[root@centos9 ~]# **buildah mount ubuntu-working-container**

/home/bobb/.local/share/containers/storage/overlay/83e3d0982907c745c6ef78d7b16118ce320ca721865e1efa68d1d5d9e52c152f/merged
[root@centos9 ~]# buildah run ubuntu-working-container -- apt-get update

buildah run ubuntu-working-container -- apt-get install -y coreutils

```
Hit:1 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu jammy InRelease
Hit:3 http://archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu jammy-backports InRelease
Reading package lists... Done
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
coreutils is already the newest version (8.32-4.1ubuntu1.3).
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
```

[root@centos9 ~]# **buildah umount ubuntu-working-container**

[root@centos9 ~]# **buildah commit ubuntu-working-container my-coreutils-image**
Getting image source signatures
Copying blob beb59027b96d skipped: already exists

```
Copying blob af95ea819889 done   |
Copying config 8b37535a8c done   |
Writing manifest to image destination
8b37535a8ca04000a6c91ed9f75a428399e34f953223af4673d00cd3078acd91

[root@centos9 ~]# podman run -it my-coreutils-image bash
root@d4abe80a2e1a:/#
```